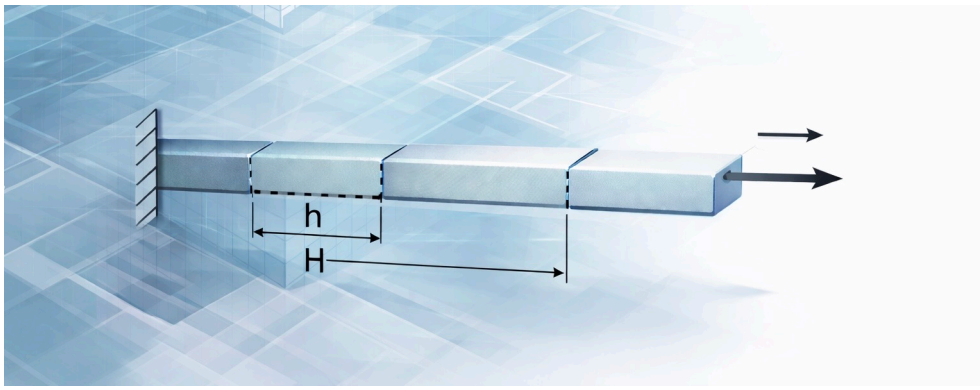
MEMDD report 2026

Study and comparison of classical DDM



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1 Introduction

As the scale of computational mechanics problems continues to grow, solving large-scale linear systems of equations directly on a single processor has become increasingly inadequate for engineering demands. Domain Decomposition Methods (DDM), as an efficient computational framework, decompose the original problem into smaller subproblems and impose compatibility conditions at the interfaces, thereby achieving iterative solution of the global problem.

Domain decomposition methods can be broadly classified into three categories. The **primal approach** starts from displacement continuity, enforcing displacement compatibility conditions at the interface to reduce the interface problem to the solution of a primal Schur complement equation; a representative method is the BDD (Balancing Domain Decomposition) method. The **dual approach** is based on interface force equilibrium, introducing Lagrange multipliers to describe interface forces and transforming the interface problem into a dual Schur complement equation, with the FETI (Finite Element Tearing and Interconnecting) method as its representative. The **mixed approach** treats both interface displacements and interface forces simultaneously, represented by the LaTIn (Large Time Increment) method, which achieves convergence by alternating iterations between local problems and global search directions.

One of the core metrics for evaluating the performance of domain decomposition methods is **numerical scalability**: while keeping the substructure-to-element size ratio h/H fixed (i.e., simultaneously increasing the number of substructures and the global problem size), the number of iterative steps to convergence should be independent of, or only weakly dependent on, the number of substructures.

The report uses a one-dimensional linear elastic bar under tension as the model problem. Consider a uniform bar of length L, cross-sectional area A, and Young's modulus E, with its left end fixed and a tensile force F_d applied at the right end. It should be noted that, in describing the implementation of each algorithm, we avoid repeating code or previously explained content, and therefore only list selected formulas as hints; detailed derivations and algorithms are provided in the appendix. The focus of the report is on the accuracy of the implementation and the explanation of certain key choices made in the code.

2 Preliminary Work

2.1 Analytical Solution

For this one-dimensional linear elastic problem, the analytical solution of the displacement field is given by:

$$u(x) = \frac{F_d}{EA}\, x \tag{2.1}$$

The strain and stress remain uniform along the length of the bar:

$$\varepsilon_{xx} = \frac{F_d}{EA}, \qquad \sigma_{xx} = \frac{F_d}{A} \tag{2.2}$$

It can be observed that both the strain and the stress remain constant along the axial direction of the bar, indicating that this problem corresponds to a typical state of uniform tension. This analytical solution will serve as the reference benchmark for the subsequent numerical results obtained by the finite element discretization and domain decomposition methods, in order to verify the computational accuracy and assess the error behavior. The detailed derivation is provided in Appendix A.1.

2.2 Finite Element Discretization

The bar is uniformly divided into n_e linear bar elements of length $h = L/n_e$, yielding $n_e + 1$ nodes in total (Figure 1). Each element employs a two-node linear displacement interpolation with a constant strain matrix, which can be integrated exactly.

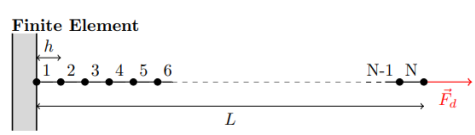


Figure 1: Schematic of the finite element discretization of the one-dimensional bar

The local stiffness matrix of element e (exact integration) is given by (derivation in Appendix A.2):

$$\boldsymbol{k}^{(e)} = \frac{EA}{h}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \tag{2.3}$$

The global stiffness matrix $\boldsymbol{K} \in \mathbb{R}^{(n_e+1)\times(n_e+1)}$ is assembled by superimposing the contributions of all elements,

resulting in a tridiagonal symmetric positive semi-definite matrix. After applying boundary conditions, the global linear system reads:

$$\boldsymbol{K}\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_{n_e+1} \end{pmatrix} = \begin{pmatrix} f_1 \\ 0 \\ \vdots \\ F_d \end{pmatrix} \tag{2.4}$$

where f_1 is the reaction force at the left constrained end, and $u_1 = 0$. This report implements and compares three methods for imposing boundary conditions: the direct elimination method, the penalty method, and the Lagrange multiplier method (detailed derivations in Appendix A.2). Since the errors of all three methods relative to the analytical solution nearly coincide, Figure 2 presents the plotted results of the three methods.

2.3 Domain Decomposition Methods

The elastic bar is partitioned into $\mathcal{S}$ subdomains, each containing N_s finite elements (of length h), $n_s = N_s + 1$ nodes, and subdomain length $H = N_s h$. Let $k_0 = EA/h$.

The nodes in each subdomain are classified as **interior nodes** (subscript i) and **interface nodes** (subscript b), and the local stiffness equations are partitioned accordingly:

$$\begin{pmatrix} K_{ii}^{(s)} & K_{ib}^{(s)} \\ K_{bi}^{(s)} & K_{bb}^{(s)} \end{pmatrix} \begin{pmatrix} u_i^{(s)} \\ u_b^{(s)} \end{pmatrix} = \begin{pmatrix} f_i^{(s)} \\ f_b^{(s)} \end{pmatrix}. \tag{2.5}$$

Eliminating the interior degrees of freedom yields the **Primal Schur complement** and right-hand side for each subdomain:

$$S_p^{(s)} = K_{bb}^{(s)} - K_{bi}^{(s)} \big(K_{ii}^{(s)}\big)^{-1} K_{ib}^{(s)} \tag{2.6}$$

$$b_p^{(s)} = f_b^{(s)} - K_{bi}^{(s)} \big(K_{ii}^{(s)}\big)^{-1} f_i^{(s)} \tag{2.7}$$

The results for each subdomain type are summarized in Table 1; detailed derivations are given in Appendix A.3.

Defining the primal assembly operator $A^{(s)}$ and concatenating to form $\mathbb{A}$ (see appendix), the global primal interface equation can be constructed as:

$$\mathbf{S}_p\, U_b = \mathbf{B}_p, \qquad \mathbf{S}_p = \mathbb{A}\, \mathbb{S}_p^{\diamondsuit}\, \mathbb{A}^T, \quad \mathbf{B}_p = \mathbb{A}\, \mathbb{B}_p^{\diamondsuit}. \tag{2.8}$$

Once the interface displacements are obtained, the interior degrees of freedom of each subdomain are recovered through a local Dirichlet problem:

$$u_i^{(s)} = \big(K_{ii}^{(s)}\big)^{-1} \Big(f_i^{(s)} - K_{ib}^{(s)}\, u_b^{(s)}\Big). \tag{2.9}$$

Table 1: Summary of Schur complements for each subdomain type

Subdomain type	$S_p^{(s)}$	$b_p^{(s)}$	Rigid body mode $R_b^{(s)}$
(1)	k_0/N_s	0	None
$(2 \cdots \mathcal{S}-1)$	$\frac{k_0}{N_s}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$	$\mathbf{0}$	$(1,1)^T$
$(\mathcal{S})$	0	F_d	1

3 Interface problem research methods

In Section 2.3 and its appendix, we constructed the Primal Schur complement $S_p^{(s)}$ for each subdomain and its assembled form

$$\mathbf{S}_p = \mathbb{A}\, \mathbb{S}_p^{\diamondsuit}\, \mathbb{A}^T.$$

The original finite element problem is thus equivalent to solving the following system at the interface:

$$\mathbf{S}_p\, U_b = \mathbf{B}_p. \tag{3.1}$$

The dimension of this interface problem is only $\mathcal{S} - 1$, which is far smaller than the original global system. Therefore, the core of domain decomposition methods lies in how to efficiently solve equation (3.1), and subsequently recover the interior degrees of freedom of each subdomain.

This section introduces three representative classes of domain decomposition methods:

- **Primal methods**: directly or iteratively solve for the interface displacements;
- **Dual methods**: enforce displacement compatibility through interface forces (Lagrange multipliers);
- **Mixed methods**: a combined framework treating both displacements and interface forces (LaTIn).

Figure 2 presents in advance a comparison of the displacement fields obtained by all methods. It can be seen that, regardless of whether the three boundary condition treatments from Section 1 or the various iterative solvers under the Primal, Dual, and Mixed frameworks are considered, all numerical solutions agree closely with the linear analytical solution $u(x) = F_d x/(ES)$, verifying the correctness of each implementation. Furthermore, Figure 12 shows that the relative errors of all five methods are close to machine precision, indicating very high numerical accuracy in

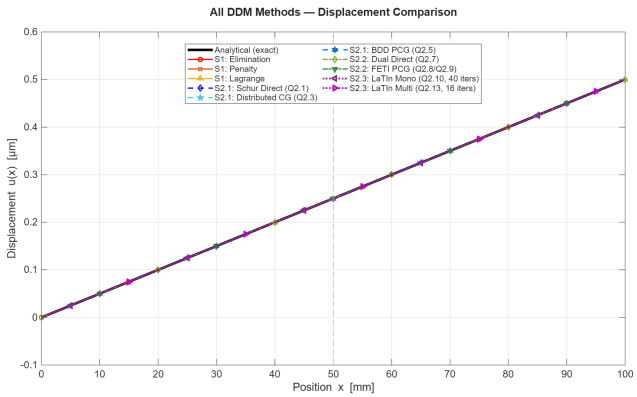


Figure 2: Comparison of displacement fields for all domain decomposition methods ($N_{\text{sub}} = 2$, $N_e = 20$).

this test; the differences among these methods primarily reflect the sensitivity of different numerical formulations to floating-point rounding errors. The subsequent sections will further analyze the convergence rates, condition number behavior, and numerical scalability of the different methods.

4 Primal Methods

4.1 Primal Direct Method

Given the interface system

$$\mathbf{S}_p\, U_b = \mathbf{B}_p,$$

the idea of the Primal direct method is to obtain the interface displacements U_b in a single solve, followed by recovery of the interior degrees of freedom for each subdomain.

Direct solution of interface displacements. Using the assembled interface system, the interface displacements are solved directly. Since $\mathbf{S}_p$ is a symmetric matrix and is positive definite for this problem, the linear system can be solved directly:

$$U_b = \mathbf{S}_p^{-1}\mathbf{B}_p.$$

In the **Matlab** implementation, this system is solved using the backslash operator.

Recovery of interior degrees of freedom. Once the interface displacements U_b are obtained, the interior degrees of freedom can be recovered for each subdomain independently. The subdomain boundary displacements are extracted using the transpose of the assembly operator:

$$u_b^{(s)} = A^{(s)T} U_b.$$

For the s-th subdomain, we have

$$k_{ii}^{(s)} u_i^{(s)} = f_i^{(s)} - k_{ib}^{(s)} u_b^{(s)},$$

and therefore

$$u_i^{(s)} = \left(k_{ii}^{(s)}\right)^{-1} \left(f_i^{(s)} - k_{ib}^{(s)} u_b^{(s)}\right).$$

It can be seen that this method reduces the original problem of size N to an interface problem of size $\mathcal{S} - 1$, after which the global solution is reconstructed through local back-substitution.

4.2 Condition Number Analysis and Scalability Test

To investigate the numerical properties of the interface system, we analyze the condition number behavior of both the global stiffness matrix K and the interface matrix $\mathbf{S}_p$. The condition number is defined as

$$\kappa(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}.$$

Since a displacement-type boundary condition (Dirichlet) is applied at the left end: $u_1 = 0$, the first degree of freedom is a known quantity rather than an unknown. Therefore, in the solution and condition number analysis, we directly eliminate this degree of freedom and retain only the submatrix $K_{FF} = K(2\!:\!end,\, 2\!:\!end)$ corresponding to the free unknowns. This elimination is fully equivalent to imposing $u_1 = 0$ in the original system, but avoids erroneous condition number estimates caused by modifying the constrained degree of freedom.

Fixed subdomain size H, refinement of the internal mesh

In this experiment, the number of subdomains $\mathcal{S}$ is held fixed while the element size h is progressively reduced, i.e., the number of elements within each subdomain is increased.

For a one-dimensional linear finite element problem, the global stiffness matrix satisfies the classical estimate (see [1]):

$$\kappa(K) = \mathcal{O}(h^{-2}).$$

Figure 3 presents the numerical results. The following observations can be made:

- The condition number of the global matrix K grows linearly as the internal mesh is refined;

- The condition number of the interface matrix $\mathbf{S}_p$ remains essentially constant.

This indicates that, once the interior degrees of freedom have been eliminated, internal mesh refinement has virtually no effect on the interface problem. In other words, even as the internal discretization within each subdomain becomes finer, the numerical properties of the interface system remain stable.

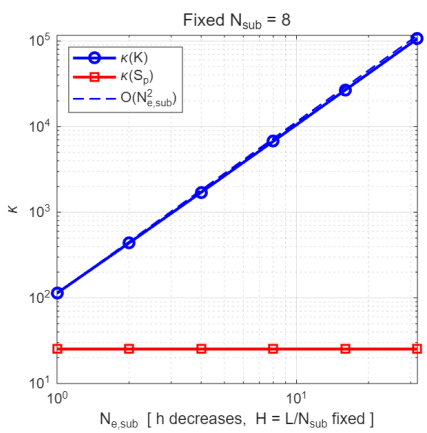


Figure 3: Condition numbers for fixed H (refinement of internal mesh).

Fixed h/H, increasing number of subdomains

In this experiment, the internal discretization ratio of each subdomain is kept fixed while the number of subdomains $\mathcal{S}$ is increased, thereby increasing the dimension of the interface unknowns.

Figure 4 shows that the condition number of the interface matrix (see [2]) satisfies

$$\kappa(\mathbf{S}_p) \sim \mathcal{O}(\mathcal{S}^2).$$

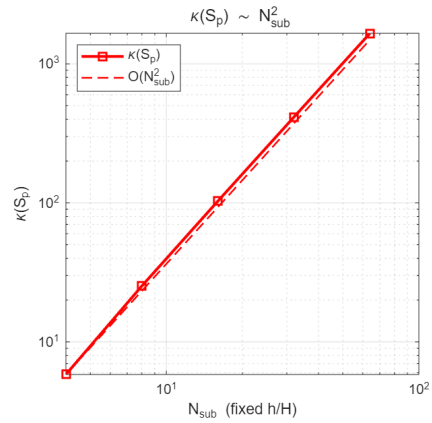


Figure 4: Condition number of $\mathbf{S}_p$ for fixed h/H (scalability test).

It can be seen from the figure that, when h/H is fixed and the number of subdomains is increased, the condition number of the interface matrix $\mathbf{S}_p$ grows rapidly. This indicates that, although the Schur complement eliminates the effect of internal mesh refinement, as the number of subdomains increases the interface system itself becomes progressively harder to solve. It can therefore be concluded that the single-scale Primal method does not possess good numerical scalability as the number of subdomains grows. As the number of interface unknowns increases, the system becomes increasingly ill-conditioned, and direct solution becomes inefficient.

4.3 Distributed Conjugate Gradient Method for the Interface Problem and Analysis

In practice, the interface matrix $\mathbf{S}_p$ is not explicitly assembled. The matrix-vector product $\mathbf{S}_p\mathbf{d}_b$ is computed through local Dirichlet problems and local matrix-vector multiplications within each subdomain, with only interface vectors exchanged during the assembly phase, so that the communication volume is $O(\mathcal{S}-1)$.

The convergence criterion is based on the normalized norm of the interface residual:

$$\eta_k = \frac{\|\mathbf{r}_{b(k)}\|}{\|\mathbf{r}_{b(0)}\|} \leq \varepsilon_{\text{tol}}, \qquad \varepsilon_{\text{tol}} = 10^{-10}. \tag{4.1}$$

The interface residual norm is equivalent to the global residual norm (proof in Appendix B.1):

$$\|\mathbf{r}_{b(k)}\|_\Gamma = \|f - Ku_{(k)}\|. \tag{4.2}$$

This criterion therefore simultaneously guarantees the accuracy of the original problem. The complete algorithmic procedure is given in Appendix B.2.

Figure 5 shows the number of iterations of the unpreconditioned distributed CG as a function of the number of subdomains $\mathcal{S}$, under fixed h/H.

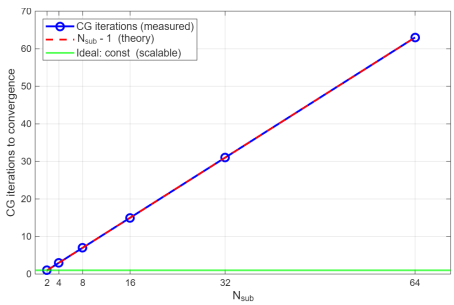


Figure 5: Number of iterations of unpreconditioned distributed CG as a function of the number of subdomains $\mathcal{S}$ (fixed h/H).

From the figure, it can be observed that the iteration count approximately satisfies

$$\text{iterations} \approx \mathcal{S} - 1,$$

i.e., it grows linearly with the dimension of the interface unknowns. Indeed, in the present one-dimensional chain-like decomposition, the interface Schur complement matrix has size $(\mathcal{S}-1)\times(\mathcal{S}-1)$, and the conjugate gradient method converges in at most n steps for an $n\times n$ symmetric positive definite matrix in the ideal case [3]. The linear relationship observed above therefore has a clear linear-algebraic explanation.

From an asymptotic perspective,

$$\text{iterations} = O(\mathcal{S}),$$

which is consistent with the result obtained in the previous subsection,

$$\kappa(\mathbf{S}_p) \sim O(\mathcal{S}^2).$$

Since the convergence rate of the conjugate gradient method is of the same order as $\sqrt{\kappa(\mathbf{S}_p)}$, the linear growth of iteration count with $\mathcal{S}$ is a theoretically expected result.

It can therefore be concluded that the unpreconditioned distributed CG method does not possess numerical scalability, as the condition number of the interface operator grows with the number of subdomains.

4.4 BDD Preconditioned Conjugate Gradient Method

The previous section showed that the iteration count of unpreconditioned distributed CG grows linearly with the number of subdomains $\mathcal{S}$, with the root cause being $\kappa(\mathbf{S}_p) \sim \mathcal{O}(\mathcal{S}^2)$. To restore numerical scalability, the Balancing Domain Decomposition (BDD) method introduces two enhancements over standard CG: (1) a Neumann preconditioner $\widetilde{\mathbf{S}}_p^{-1}$ to compress high-frequency errors in the spectrum; (2) a coarse space projection P to eliminate low-frequency (rigid body mode) errors.

Method Overview

Floating substructures and rigid body modes. The left-end subdomain ($s=1$) contains the Dirichlet boundary condition, while the remaining subdomains $s=2,\dots,\mathcal{S}$ are *floating substructures* whose local stiffness matrices $K^{(s)}$ are singular (and cannot be directly inverted under Neumann boundary conditions). The null space of each floating substructure is spanned by its rigid body modes; for the 1D scalar displacement problem, this null space is the one-dimensional space of constant translations.

Coarse space matrix G. Let $G \in \mathbb{R}^{n_\Gamma \times n_G}$, where each column is the restriction of a rigid body mode of a floating substructure to the interface. In this problem there are $\mathcal{S}-1$ floating substructures, each contributing one column, so $n_G = \mathcal{S}-1 = n_\Gamma$ (the number of interface nodes equals the number of floating substructures).

Projection operator. Define the orthogonal projection operator on the interface as

$$P = I - G\big(G^T\mathbf{S}_pG\big)^{-1}G^T\mathbf{S}_p, \tag{4.3}$$

which projects any interface vector onto the subspace $\mathbf{S}_p$-orthogonal to all rigid body modes, so that the operator $P^T\mathbf{S}_pP$ to be solved by the subsequent CG is well-defined. The initial interface displacement is given by:

$$\mathbf{u}_0 = G\big(G^T\mathbf{S}_pG\big)^{-1}G^T\mathbf{b}_p. \tag{4.4}$$

Neumann preconditioner. The Neumann preconditioner $\widetilde{\mathbf{S}}_p^{-1}$ uses the local Neumann Schur complement of

each subdomain to process the interface residual locally:

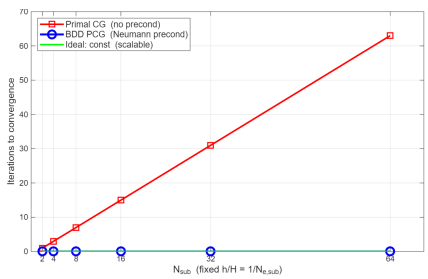
$$\widetilde{\mathbf{S}}_p = \mathbb{A}\,\widetilde{\mathbf{S}}_p^{\Diamond}\,\mathbb{A}^T, \qquad \widetilde{S}_p^{(s)} = K_{bb}^{(s)} - K_{bi}^{(s)}\big(K_{ii}^{(s)}\big)^{\dagger} K_{ib}^{(s)}, \quad (4.5)$$

where $(\cdot)^{\dagger}$ denotes the Moore–Penrose pseudoinverse.

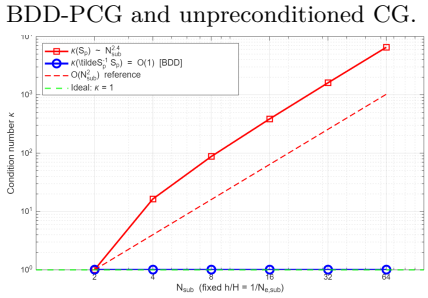
The complete algorithmic procedure is given in Appendix C.1.

Numerical Results and Analysis

Scalability test (fixed $h/H = 1/N_{e,\mathrm{sub}}$). Figure 6 presents, for fixed h/H with varying number of subdomains $\mathcal{S}$, a comparison of the iteration counts between BDD-PCG and unpreconditioned CG, as well as the corresponding comparison of condition numbers.



(a) Iteration count comparison between BDD-PCG and unpreconditioned CG.



(b) Condition number comparison before and after projection.

Figure 6: Iteration count comparison between BDD-PCG and unpreconditioned CG, and condition number comparison before and after projection.

The following observations can be made from the figure:

- The iteration count of unpreconditioned CG grows linearly with $\mathcal{S}$, confirming the theoretical prediction $\kappa(\mathbf{S}_p) \sim \mathcal{O}(\mathcal{S}^2)$;
- The iteration count of BDD-PCG is **0**, independent of the number of subdomains.

Theoretical explanation for zero iterations of BDD in this 1D problem. This phenomenon is not an implementation error, but rather a structural degeneracy of the 1D chain decomposition. The key result is as follows:

For the 1D chain decomposition, the matrix G forms a unit bidiagonal lower triangular matrix with $\det(G) = 1$, so G is invertible, and the coarse space completely spans the interface space $\mathbb{R}^{n_\Gamma}$. Consequently the projection operator $P \equiv 0$, and the initial solution $\mathbf{u}_0$ (equation (4.4)) already satisfies the interface equation exactly, so CG requires no iterations.

$$\|\mathbf{b}_p - \mathbf{S}_p\mathbf{u}_0\| \;\sim\; \mathcal{O}(10^{-12}), \quad (4.6)$$

i.e., the residual has already reached machine precision, and the iteration count is therefore recorded as 0.

From the condition number perspective, since the projected operator is identically zero, measuring a degenerate matrix with the conventional $\kappa = \lambda_{\max}/\lambda_{\min}$ is no longer meaningful; the value $\kappa(P^T\mathbf{S}_p P) = 1$ recorded in the figure reflects precisely this degeneracy.

5 Dual Methods

Unlike the Primal method, which takes the interface **displacements** as the primary unknowns, the Dual method takes the interface **forces** (Lagrange multipliers λ_b) as the primary unknowns. By introducing interface forces $\underline{F}_{EE'}$ at each interface $\Gamma_{EE'}$ to enforce displacement compatibility ($\underline{F}_{EE'} = \lambda_b$), the global problem is decomposed into independent local subproblems (the FETI method).

The Dual form of the interface problem yields the following saddle-point system (see Appendix D.1):

$$\boxed{\begin{pmatrix} S_d & G \\ G^T & 0 \end{pmatrix}\begin{pmatrix} \lambda_b \\ \alpha_b^{\Diamond} \end{pmatrix} = \begin{pmatrix} -b_d \\ -e^{\Diamond} \end{pmatrix}} \quad (5.1)$$

where the operators are defined as

$$S_d = \mathbb{A}^{\Diamond} S_d^{\Diamond} \mathbb{A}^{\Diamond T}, \quad (5.2)$$

$$G = \mathbb{A}^{\Diamond} R_b^{\Diamond}, \quad (5.3)$$

$$b_d = \mathbb{A}^{\Diamond} S_d^{\Diamond} b_p^{\Diamond}, \quad (5.4)$$

$$e^{\Diamond} = R_b^{\Diamond T} b_p^{\Diamond}. \quad (5.5)$$

Here S_d is the **dual Schur complement**. The complete derivation of system (5.1) is given in Appendix D.1.2.

5.1 Direct Solution of the Dual Interface Problem

After explicitly assembling the operators in equations (D.9)–(D.12), the saddle-point linear system (5.1) is solved directly to obtain λ_b and $\alpha_b^\diamondsuit$. The subdomain boundary displacements $u_b^{(s)}$ are then recovered via equation (D.6), followed by recovery of the interior displacements $u_i^{(s)}$ from the local interior equations.

Physical interpretation of interface forces. Under uniform tension, the Lagrange multipliers at each interface should equal the internal interface force, and the results again confirm the description given earlier:

$$\lambda_j = F_d, \quad j = 1, \ldots, \mathcal{S} - 1. \tag{5.6}$$

The complete displacement recovery algorithm is given in Appendix D.2.

5.2 FETI Method —Distributed Projected Preconditioned Conjugate Gradient

As with the Primal direct method, the Dual direct method requires explicit assembly of S_d, which is not feasible in practical parallel computations. The FETI method instead employs a **projected preconditioned conjugate gradient** (Projected PCG) to iteratively solve for λ_b in the saddle-point system (5.1), with **only interface vectors exchanged** between subdomains.

Constrained subspace and projection operator. The constraint row of the saddle-point system (5.1) requires $G^T\lambda_b = -e^\diamondsuit$ (the rigid body equilibrium condition for each floating substructure). To this end, the projection operator

$$P = I - G\big(G^TG\big)^{-1}G^T \tag{5.7}$$

is introduced so that the iterates always satisfy $G^TPv = 0$, i.e., the iterative increments remain in the constrained subspace $\ker(G^T)$. The initial guess $\lambda_b^{(0)}$ is directly determined from the constraint condition:

$$\lambda_b^{(0)} = -G\big(G^TG\big)^{-1}e^\diamondsuit, \tag{5.8}$$

and it can be verified that $G^T\lambda_b^{(0)} = -e^\diamondsuit$, i.e., the admissibility condition is satisfied.

Dirichlet preconditioner. Unlike the BDD method, which uses a Neumann preconditioner, the FETI method employs a Dirichlet preconditioner:

$$\widetilde{S}_d^{-1} = \mathbb{A}^\diamondsuit S_p^\diamondsuit \mathbb{A}^{\diamondsuit T}, \tag{5.9}$$

i.e., the stiffness matrix assembled from the primal Schur complements $S_p^{(s)}$ of each subdomain serves as an approximate inverse of S_d, also computed in a distributed manner without assembling the global matrix. (The implementation follows the distributed algebraic formulation, although in this project the operations were executed sequentially on a single machine.)

Convergence criterion and post-processing. The convergence criterion is based on the normalized norm of the projected residual:

$$\eta_k = \frac{\|P\,r_{(k)}\|}{\|P\,r_{(0)}\|} \le \varepsilon_{\text{tol}} = 10^{-10}, \tag{5.10}$$

where $r_{(k)} = S_d\lambda_{(k)} + b_d$. Upon convergence, the rigid body mode amplitudes $\alpha_b^\diamondsuit$ are determined by a post-processing step:

$$\alpha_b^\diamondsuit = \big(G^TG\big)^{-1}G^T\big(-b_d - S_d\lambda_b^{(m)}\big). \tag{5.11}$$

The complete algorithmic pseudocode is given in Appendix D.4 (see also [4] for details).

Numerical results and scalability. For the present 1D uniform tension problem, the initial guess (5.8) already satisfies the interface equation exactly, so PCG **converges without any iterations**. This is a structural degeneracy of the 1D chain decomposition, analogous to the analysis of the BDD method in Section 4.4.

Figure 7 presents a comparison of the FETI iteration count and the unpreconditioned CG iteration count as functions of the number of subdomains $\mathcal{S}$, under fixed h/H. The FETI iteration count is constantly 0 (red line), while the unpreconditioned CG iteration count grows linearly with $\mathcal{S}$ (blue line), directly demonstrating the **numerical scalability** of the FETI method.

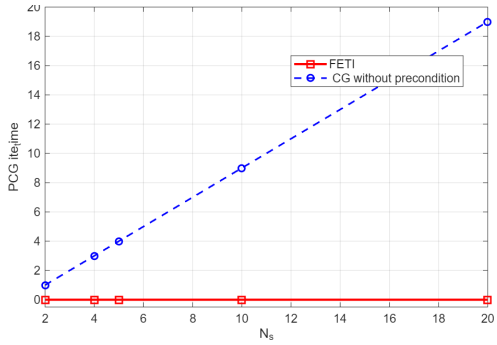


Figure 7: Comparison of iteration counts between FETI (Dirichlet preconditioner) and unpreconditioned CG as a function of the number of subdomains $\mathcal{S}$, under fixed h/H.

5.3 Rigid Body Mode Recovery and Displacement Field Reconstruction

After FETI-PCG converges, the interface forces $\lambda_b^{(m)}$ are obtained, but the displacement solution of each floating substructure remains non-unique. The particular solution given by the pseudoinverse is $\tilde{u}^{(s)} = S_d^{(s)}(b_p^{(s)} + \lambda_b^{(s)})$.

The complete displacement solution requires adding the rigid body mode correction:

$$u_b^{(s)} = \tilde{u}_b^{(s)} + R_b^{(s)}\,\alpha^{(s)}, \tag{5.12}$$

where the amplitudes $\alpha^{(s)}$ are directly obtained from equation (5.11).

Physical interpretation. For the 1D problem, $R_b^{(s)} = \mathbf{1}/\|\mathbf{1}\|$ (constant translation mode), and $\alpha^{(s)}$ represents the rigid body translation of each floating subdomain. The particular solution $\tilde{u}^{(s)}$, determined by the pseudoinverse, reflects only the elastic deformation contribution; the term $R_b^{(s)}\alpha^{(s)}$ then "translates" each subdomain to the correct position such that interface displacement continuity is satisfied.

For the substructure containing the Dirichlet boundary condition (substructure 1 in this problem), $S_p^{(1)}$ is positive definite and invertible, so no rigid body mode exists and the particular solution is the complete solution: $\tilde{u}^{(1)} = u^{(1)}$.

Numerical results. Figure 8 shows a comparison between the particular solution (dashed lines) and the complete solution (solid lines) for the two substructures. The particular solution of substructure 2 starts from 0 and grows linearly, producing a jump at the interface ($x = 50\,\text{mm}$) relative to substructure 1; after adding the rigid body translation $\alpha^{(2)} \approx 0.25\,\mu\text{m}$, the complete solution is continuous at the interface, and the relative L^2 error between the global displacement field and the analytical solution $u(x) = F_d x/(ES)$ is 2.86×10^{-15}.

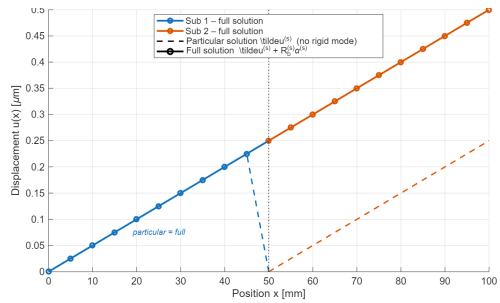


Figure 8: Rigid body mode recovery: comparison between the particular solution (dashed lines) and the complete solution (solid lines) for each substructure.

6 Mixed Methods

The Mixed method treats both interface displacements and interface forces as unknowns simultaneously. By introducing search direction operators k^+ and k^-, the global problem is decomposed into alternating linear stages (global) and local stages (each substructure independently). This section employs the single-scale Mixed DDM based on the LATIN algorithm (Algorithm 5) (see [5] for details).

Problem Formulation and Variable Definitions

The structural domain is decomposed into substructures $E \in \mathbf{E}$ and their interfaces $\Gamma_{EE'}$. The state of each substructure is described by a displacement field u_E (with its interface trace W_E) and a force field F_E.

At each interface $\Gamma_{EE'}$, **interface variables** $(\widehat{W}_{EE'}, \widehat{F}_{EE'})$ are introduced, satisfying:

$$\widehat{W}_{EE'} = \widehat{W}_{E'E}, \qquad \widehat{F}_{EE'} + \widehat{F}_{E'E} = 0, \tag{6.1}$$

i.e., the interface variables satisfy displacement continuity and force equilibrium.

Search Directions and Two-Stage Iteration

The LATIN algorithm connects the substructure solution space $\mathcal{A}_d$ to the interface Γ through two **search di-**

rections E^+ (slope k^+) and E^- (slope k^-), alternating between:

Linear stage. Given the interface quantities from the previous step $(\widehat{W}_E^{(n-1/2)}, \widehat{F}_E^{(n-1/2)})$, a local problem with boundary conditions is solved for each substructure:

$$\left([K_E] + [k_E^-]\right)[u_E^{(n)}] = [\widehat{F}_{E,d}], \tag{6.2}$$

where the right-hand side includes the interface forces from the previous step and the search direction correction term. The forces are then updated along the search direction E^-:

$$F_E^{(n)} = \widehat{F}_E^{(n-1/2)} - k^-\left(W_E^{(n)} - \widehat{W}_E^{(n-1/2)}\right). \tag{6.3}$$

Local stage. For each interface $\Gamma_{EE'}$, starting from the current substructure solution, the interface quantities are solved along the search direction E^+:

$$\widehat{W}_{EE'}^{(n+1/2)} = \frac{1}{2}\left(W_{EE'}^{(n)} + W_{E'E}^{(n)}\right) - \frac{(k^+)^{-1}}{2}\left(F_{EE'}^{(n)} + F_{E'E}^{(n)}\right), \tag{6.4}$$

$$\widehat{F}_{EE'}^{(n+1/2)} = F_{EE'}^{(n)} + k^+\left(\widehat{W}_{EE'}^{(n+1/2)} - W_{EE'}^{(n)}\right). \tag{6.5}$$

It can be verified that equation (6.1) is automatically satisfied after this step.

1D Discretization of the Search Direction Operators

For the present 1D bar problem, both k^+ and k^- are scalars (one degree of freedom per interface node). In the linear stage, the contribution of k^- is added to the local stiffness matrix of each substructure, i.e., an additional stiffness is introduced at the interface degrees of freedom:

$$[K_E^{\text{aug}}] = [K_E] + k^- \cdot [I_b], \tag{6.6}$$

where $[I_b]$ is the identity matrix with nonzero entries only at the interface degrees of freedom.

6.1 Single-Scale LATIN: Solution Results

Convergence criterion. The error indicator from the reference (see Appendix E.2) is adopted:

$$\eta^2 = \frac{\sum_{E\in\mathbf{E}}\left[\|W_E - \widehat{W}_E\|_{W,k}^2 + \|F_E - \widehat{F}_E\|_{F,k}^2\right]}{\sum_{E\in\mathbf{E}}\left[\|W_E\|_{W,k}^2 + \|\widehat{W}_E\|_{W,k}^2 + \|F_E\|_{F,k}^2 + \|\widehat{F}_E\|_{F,k}^2\right]}, \tag{6.7}$$

which measures the conjugacy between the substructure solution and the interface solution along the search directions; convergence is achieved when $\eta \to 0$.

Numerical results. Taking search direction parameters $k^+ = k^- = k_0 = EA/h$ (of the same order as the element stiffness), with convergence threshold $\varepsilon_{\text{tol}} = 10^{-10}$, the relative L^2 error between the numerical displacement field and the analytical solution $u(x) = F_d x/(EA)$ is at machine precision level, verifying the correctness of the implementation.

The complete algorithm is given in Appendix E.1.

6.2 Optimization of the Search Direction Parameter

Optimal value analysis. A numerical parameter sweep (Figure 9, left) shows that the iteration count is minimized at $k^{\text{opt}} = EA/H$. This result has a clear mechanical interpretation: for the present 1D bar problem, the primal Schur complement at the substructure interface is

$$S_p^{(s)} = \frac{EA}{H}, \tag{6.8}$$

i.e., after condensing the interior degrees of freedom, the equivalent stiffness at the interface is exactly EA/H. The search direction operators $k^\pm$ act as interface springs in the algorithm: when $k^\pm = S_p^{(s)}$, the two search directions E^+ and E^- in the linear and local stages are exactly mutually orthogonal (conjugate), so that each iterative correction points precisely toward the solution, and convergence is theoretically achieved in a single step.

1D degeneracy. Since all substructures of the 1D elastic bar share the same uniform stiffness and $S_p^{(s)} = EA/H$ is identical for all s, the optimal k is unique and can be determined exactly.

Effect of deviating from the optimal value. As seen in Figure 9: for $k < k^{\text{opt}}$, the convergence rate degrades slowly as k decreases (low-stiffness direction); for $k > k^{\text{opt}}$, the convergence rate deteriorates rapidly as k increases (asymmetric high-stiffness direction). This behavior is consistent with the sensitivity of penalty parameter selection in augmented Lagrangian methods.

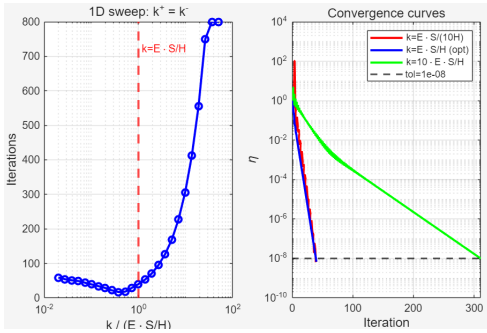


Figure 9: LaTIn search direction parameter study ($N_{\text{sub}} = 2$, $N_{e,\text{sub}} = 10$). Left: iteration count as a function of $k/(EA/H)$, with the optimal value at $k = EA/H$; right: convergence history for three representative values of k, showing that the optimal k converges fastest.

6.3 Numerical Scalability of Single-Scale LATIN

Figure 10 shows the LATIN iteration count as a function of the number of subdomains N_{sub} under fixed h/H (i.e., fixed $N_{e,\text{sub}}$). The iteration count grows noticeably with N_{sub}, deviating significantly from the ideal flat baseline, indicating that **single-scale LATIN does not possess numerical scalability**.

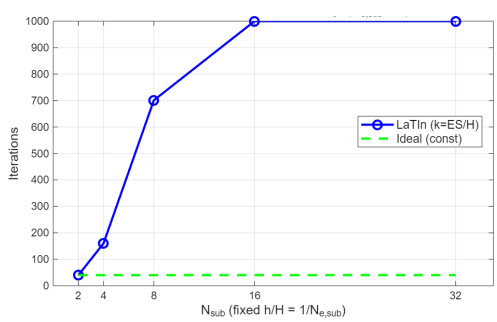


Figure 10: Scalability analysis of single-scale LaTIn ($k = EA/H$, fixed $h/H = 1/N_{e,\text{sub}}$).

The fundamental reason is that the interface iterations of single-scale LATIN cannot propagate global (low-frequency) information: in each iteration, the interface correction is propagated only between neighboring substructures, and the number of iterations required for information to traverse the entire structure is proportional to the number of subdomains. This behavior is consistent with single-scale dual/primal methods such as BDD and FETI, all of which require the introduction of a coarse-scale (macroscopic) problem to restore scalability.

6.4 Multiscale LATIN: Macroscopic Problem

To restore scalability, a coarse (macroscopic) problem is introduced on top of single-scale LATIN, yielding the multiscale LATIN Mixed DDM (Algorithm 6, see Appendix E.4).

The macroscopic problem extracts global stiffness information by **homogenizing** the interface operator: for each substructure E, the homogenization operator $[\mathbf{L}_E^F]$ is computed and globally assembled into the macroscopic stiffness matrix $[\mathbf{L}^F]_{e^M}$. An additional small-scale macroscopic system is solved in the linear stage to accelerate the propagation of low-frequency errors:

$$[\mathbf{L}^F]_{e^M}\,[\widetilde{W}^M]_{e^M} = [\widehat{F}_d^M]_{e^M} + [F_d^M]_{e^M}. \tag{6.9}$$

1D degenerate case. For the present 1D uniform bar problem, each interface consists of a single point (zero measure), so the macroscopic problem degenerates to a scalar equation. The homogenization operator L_E^F equals the primal Schur complement of the substructure, $S_p^{(s)} = EA/H$, and the assembled macroscopic system (6.9) is equivalent to the global static equation, yielding the exact solution in a single step. Consequently the microscopic correction is zero, and the multiscale method **converges in one step** in this degenerate case, at which point the method is fully scalable (see [6] for details).

6.5 Numerical Scalability of Multiscale LATIN

Figure 11 compares the iteration counts of single-scale and multiscale LATIN as functions of N_{sub}. After introducing the macroscopic problem, the iteration count barely grows with the number of subdomains (the red line remains close to the ideal green horizontal baseline), confirming that the method possesses numerical scalability.

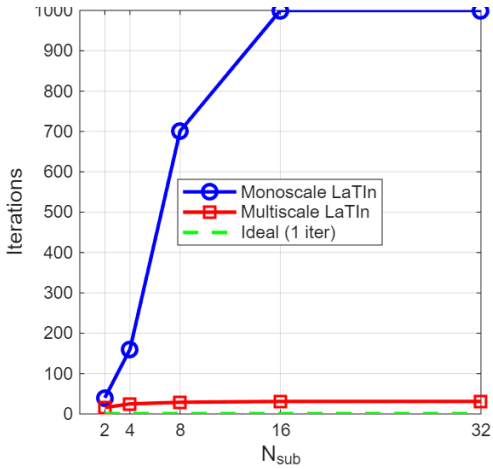


Figure 11: Scalability comparison between single-scale and multiscale LaTIn ($k = EA/H$, fixed h/H).

The physical explanation of this result follows directly from the previous subsection: the macroscopic problem propagates global low-frequency information in a single step, eliminating the bottleneck of cross-subdomain information propagation in single-scale iterations. For the present 1D uniform problem, the homogenization operator is exactly equal to the substructure Schur complement, the macroscopic system yields the exact macroscopic solution in one step, and the microscopic corrections are already sufficiently small; consequently the total iteration count of the multiscale method is independent of N_{sub}.

7 Conclusion

This project systematically implements and compares three classes of classical domain decomposition methods on the example of a 1D homogeneous bar under tension. **The Primal method (BDD)** takes the interface displacements as unknowns and solves the primal Schur complement system via a Neumann-preconditioned PCG. Numerical experiments show that the condition number after preconditioning is reduced to $\mathcal{O}(1)$; however, the monoscale method still lacks numerical scalability. **The Dual method (FETI)** takes the interface forces as unknowns and solves the dual Schur complement system via a projected PCG with a Dirichlet preconditioner. In the present 1D homogeneous problem, the initial guess already satisfies the interface equations, so PCG requires no iterations; the rigid body modes are recovered exactly through a post-processing step, and the displacement field is continuous and matches the analytical solution to machine precision. **The Mixed method (LATIN)** takes both interface displacements and forces as unknowns, and alternately solves the substructure local problems and the interface problem through a two-stage iterative scheme. The optimal value of the search direction parameter, $k^{\text{opt}} = EA/H$, equals the primal Schur complement of the substructure; at this value the two search directions are conjugate and convergence is fastest. The monoscale LATIN likewise lacks numerical scalability; upon introduction of a macro (homogenized) problem, the number of iterations of the multiscale LATIN becomes approximately independent of the number of subdomains, thereby recovering scalability. In the present degenerate 1D setting, all three methods exhibit structural simplifications (zero or very few iterations), and the true significance of their numerical scalability can only be fully appreciated in 2D/3D heterogeneous problems.

A Preliminary Work

A.1 Derivation of the Analytical Solution

Consider a homogeneous elastic bar of length L, cross-sectional area S, and Young's modulus E. The bar is fixed at the left end, while a tensile force F_d is applied at the right end. Body forces are neglected.

Equilibrium equation. Under the assumption of no body forces, the conservation of linear momentum reduces to:

$$\nabla \cdot \boldsymbol{\sigma} = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}\sigma_{xx}}{\mathrm{d}x} = 0 \tag{A.1}$$

Hence σ_{xx} is a constant C_1.

Force boundary condition. At the right end $x = L$, the resultant force acting on the cross-section gives:

$$\sigma_{xx}(x = L) \cdot S = F_d \quad \Longrightarrow \quad C_1 = \frac{F_d}{S} \tag{A.2}$$

Therefore:

$$\sigma_{xx}(x) = \frac{F_d}{S} \tag{A.3}$$

Kinematic relation. Under the assumption of small deformations, the strain tensor is $\boldsymbol{\varepsilon} = \frac{1}{2}\left(\nabla\vec{u} + (\nabla\vec{u})^T\right)$, which reduces in the one-dimensional case to:

$$\varepsilon_{xx} = \frac{\mathrm{d}u_x}{\mathrm{d}x} \tag{A.4}$$

Constitutive relation. For a one-dimensional isotropic linear elastic material, Hooke's law gives:

$$\sigma_{xx} = E\,\varepsilon_{xx} \quad \Longrightarrow \quad \frac{\mathrm{d}u_x}{\mathrm{d}x} = \frac{F_d}{ES} \tag{A.5}$$

Integration and displacement boundary condition. Integrating the above expression with respect to x:

$$u_x(x) = \frac{F_d}{ES}\,x + C_2 \tag{A.6}$$

Applying the fixed-end condition $u_x(x = 0) = 0$ yields $C_2 = 0$.

Final result.

$$\boxed{u_x(x) = \frac{F_d}{ES}\,x, \qquad \varepsilon_{xx} = \frac{F_d}{ES}, \qquad \sigma_{xx} = \frac{F_d}{S}} \tag{A.7}$$

In particular, the displacement at the right end is $u_x(L) = \dfrac{F_d L}{ES}$, consistent with the overall axial flexibility of the bar.

A.2 Derivation of the Finite Element Discretization

A.2.1 Problem Discretization

The bar of length L is uniformly divided into N elements of length $h = L/N$, yielding $n = N + 1$ nodes in total. Two-node linear displacement interpolation elements are employed.

A.2.2 Shape Functions and Strain Matrix

For element e, the coordinates of the two end nodes are x_1 and x_2 ($x_2 - x_1 = h$), with nodal displacements d_1 and d_2. Assuming a linear displacement field within the element:

$$w(x) = \varrho_0 + \varrho_1 x \tag{A.8}$$

Solving for ϱ_0 and ϱ_1 from the nodal conditions $w(x_1) = d_1$ and $w(x_2) = d_2$ gives:

$$w(x) = \underbrace{\left(1 - \frac{x}{h}\right)}_{N_1} d_1 + \underbrace{\frac{x}{h}}_{N_2} d_2 = \boldsymbol{N}\boldsymbol{d}^{(e)} \tag{A.9}$$

where the shape function matrix is $\boldsymbol{N} = \left[\dfrac{x_2 - x}{x_2 - x_1}, \dfrac{x - x_1}{x_2 - x_1}\right]$. In the one-dimensional case, the strain matrix is:

$$\boldsymbol{B} = \frac{\mathrm{d}\boldsymbol{N}}{\mathrm{d}x} = \frac{1}{h}\begin{bmatrix} -1 & 1 \end{bmatrix} \tag{A.10}$$

The strain and stress are therefore:

$$\varepsilon = \boldsymbol{B}\boldsymbol{d}^{(e)} = \frac{d_2 - d_1}{h}, \qquad \sigma = E\varepsilon = E\,\frac{d_2 - d_1}{h} \tag{A.11}$$

A.2.3 Derivation of the Element Stiffness Matrix

Neglecting body forces, the element potential energy functional for an admissible displacement $w(x)$ is:

$$J(w) = \frac{1}{2}\int_\Omega \varepsilon : E : \varepsilon\,\mathrm{d}\Omega - \int_{\partial\Omega_F} F_d \cdot w\,\mathrm{d}\Gamma \tag{A.12}$$

Substituting the approximation, for elements 1 to $N-1$ (no external forces):

$$J_{1\ldots N-1}(\boldsymbol{d}^{(e)}) = \frac{1}{2}\,\boldsymbol{d}^{(e)^T}\left[\int_{x_1}^{x_2} \boldsymbol{B}^T E \boldsymbol{B}\,S\,\mathrm{d}x\right]\boldsymbol{d}^{(e)} \tag{A.13}$$

For the last element N, a concentrated force F_d is applied at the right-end node (introduced via a Dirac distribution):

$$\begin{aligned}\int_{\partial\Omega_F} F_d \cdot w\,\mathrm{d}\Gamma &= \int_{x_1}^{x_2} F_d \cdot w(x)\,\delta(x - x_2)\,\mathrm{d}x \\ &= F_d \cdot w(x_2) \\ &= F_d \cdot d_2\end{aligned} \tag{A.14}$$

Hence:

$$J_N(\boldsymbol{d}^{(e)}) = \frac{1}{2}\,\boldsymbol{d}^{(e)^T}\left[\int_{x_1}^{x_2} \boldsymbol{B}^T E \boldsymbol{B}\, S\,\mathrm{d}x\right]\boldsymbol{d}^{(e)} - F_d \cdot d_2 \tag{A.15}$$

The element stiffness matrix is computed as:

$$\boldsymbol{k}^{(e)} = \int_{x_1}^{x_2} \boldsymbol{B}^T E \boldsymbol{B}\, S\,\mathrm{d}x = \boldsymbol{B}^T E \boldsymbol{B}\, S \cdot h = \frac{ES}{h}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \tag{A.16}$$

Applying the principle of minimum potential energy $\dfrac{\partial J}{\partial \boldsymbol{d}^{(e)}} = \boldsymbol{0}$ to each element yields the element equilibrium equations:

- Element 1: $\boldsymbol{k}^{(e)}\boldsymbol{d}^{(e)} = \begin{pmatrix} f_1 \\ 0 \end{pmatrix}$ (f_1 is the reaction force at the left constrained end)
- Elements 2 to $N-1$: $\boldsymbol{k}^{(e)}\boldsymbol{d}^{(e)} = \boldsymbol{0}$
- Element N: $\boldsymbol{k}^{(e)}\boldsymbol{d}^{(e)} = \begin{pmatrix} 0 \\ F_d \end{pmatrix}$

A.2.4 Global Assembly

By superimposing element contributions, the global stiffness matrix $\boldsymbol{K} \in \mathbb{R}^{n\times n}$ is a tridiagonal matrix:

$$\boldsymbol{K} = \frac{ES}{h}\begin{bmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 1 \end{bmatrix}_{n\times n} \tag{A.17}$$

After applying the boundary condition $u_1 = 0$ and the right-end load F_d, the complete global system reads:

$$\frac{ES}{h}\begin{bmatrix} 1 & -1 & & & \\ -1 & 2 & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & 2 & -1 \\ & & & -1 & 1 \end{bmatrix}_{n\times n}\begin{pmatrix} 0 \\ u_2 \\ \vdots \\ u_{n-1} \\ u_n \end{pmatrix} = \begin{pmatrix} f_1 \\ 0 \\ \vdots \\ 0 \\ F_d \end{pmatrix} \tag{A.18}$$

A.2.5 Imposition of Boundary Conditions

A.2.6 Direct Elimination Method

Since $u_1 = 0$ is known, the first row and first column of the global system are deleted directly, yielding a reduced system of size $(n-1)\times(n-1)$:

$$\frac{ES}{h}\begin{bmatrix} 2 & -1 & & \\ -1 & \ddots & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 1 \end{bmatrix}_{(n-1)\times(n-1)}\begin{pmatrix} u_2 \\ \vdots \\ u_{n-1} \\ u_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ F_d \end{pmatrix} \tag{A.19}$$

This system can be solved directly by inversion or an iterative process. The left-end reaction force is recovered from the first row of the original equation: $f_1 = -\dfrac{ES}{h}\,u_2$.

A.2.7 Penalty Method

Without modifying the system dimension, a penalty parameter $\gamma\dfrac{ES}{h}$ (with $\gamma = 10^4$) is added to the $(1,1)$ entry of the global stiffness matrix, while the first entry of the right-hand side is set to $f_1 + \gamma\dfrac{ES}{h}\cdot u_1^*$ ($u_1^* = 0$):

$$\frac{ES}{h}\begin{bmatrix} \gamma+1 & -1 & & & \\ -1 & 2 & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & 2 & -1 \\ & & & -1 & 1 \end{bmatrix}_{n\times n}\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_{n-1} \\ u_n \end{pmatrix} = \begin{pmatrix} f_1 + \gamma\frac{ES}{h}u_1^* \\ 0 \\ \vdots \\ 0 \\ F_d \end{pmatrix} \tag{A.20}$$

Since γ is sufficiently large, the system enforces $u_1 \approx u_1^* = 0$, thereby approximately satisfying the fixed-end condition. After solving for $u_{1,\text{approx}}, \ldots, u_{n,\text{approx}}$, the left-end reaction force is recovered as:

$$f_1 = -\gamma\,\frac{ES}{h}\,u_{1,\text{approx}} \tag{A.21}$$

This method does not alter the system dimension and is simple to implement, but introduces a condition number deterioration proportional to γ, with accuracy of order $O(1/\gamma)$.

A.2.8 Lagrange Multiplier Method

A Lagrange multiplier φ is introduced to strictly enforce the constraint $u_1 = 0$, extending the system to size

$(n+1)\times(n+1)$:

$$\frac{ES}{h}\begin{bmatrix} 1 & -1 & & & 1 \\ -1 & 2 & \ddots & & 0 \\ & \ddots & \ddots & -1 & \vdots \\ & & -1 & 1 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{bmatrix}_{(n+1)\times(n+1)} \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \\ \varphi \end{pmatrix} = \begin{pmatrix} f_1+\varphi \\ 0 \\ \vdots \\ F_d \\ 0 \end{pmatrix} \tag{A.22}$$

The last row is the constraint equation $u_1 = 0$, which exactly imposes the fixed-end condition. Setting the first entry of the right-hand side to $f_1 + \varphi = 0$, the right-hand side is fully known and the system can be solved directly. The left-end reaction force is recovered by $f_1 = -\varphi$. Unlike the penalty method, this approach introduces no approximation error, but increases the system size by one and renders the global matrix no longer positive definite.

A.3 Detailed Derivation of Block Matrices and Schur Complements

A.3.1 Element Stiffness Matrix and Subdomain Assembly

The element stiffness matrix for a two-node linear element of length h is:

$$K_e = k_0 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad k_0 = \frac{ES}{h}. \tag{A.23}$$

The local stiffness matrix is assembled following the procedure described in Appendix A.2.

A.3.2 Block Partition and Schur Complement for Subdomain (1)

The first node of subdomain (1) satisfies the boundary condition $u_1^{(1)} = 0$, and the interface node is the **last node** (local index n_s).

Original system

$$K^{(1)}u^{(1)} = f^{(1)}, \quad u^{(1)} = \begin{pmatrix} 0 \\ u_2 \\ \vdots \\ u_{n_s-1} \\ u_{n_s} \end{pmatrix}, \quad f^{(1)} = \begin{pmatrix} f_1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}. \tag{A.24}$$

Direct elimination of the first row and column

Setting $u_1^{(1)} = 0$ and eliminating the first row and column directly yields the reduced matrix $K_{\text{trim}}^{(1)}$ of size $(n_s-1)\times(n_s-1)$:

$$K_{\text{trim}}^{(1)} = k_0 \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & \ddots & \ddots & \ddots & \vdots \\ 0 & \ddots & \ddots & -1 & 0 \\ \vdots & & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 1 \end{pmatrix}_{(n_s-1)\times(n_s-1)}. \tag{A.25}$$

Interior/interface partition

In the reduced system, the interior degrees of freedom are nodes 2 to $n_s - 1$ (a total of $n_s - 2$), and the interface degree of freedom is node n_s (a total of 1). The subblocks are:

$$K_{ii}^{(1)} = k_0 \begin{pmatrix} 2 & -1 & & \\ -1 & \ddots & \ddots & \\ & \ddots & 2 & -1 \\ & & -1 & 2 \end{pmatrix}_{(n_s-2)\times(n_s-2)}, \tag{A.26}$$

$$K_{ib}^{(1)} = k_0 \begin{pmatrix} 0 \\ \vdots \\ 0 \\ -1 \end{pmatrix}_{(n_s-2)\times 1}, \tag{A.27}$$

$$K_{bi}^{(1)} = k_0 \begin{pmatrix} 0 & \cdots & 0 & -1 \end{pmatrix}_{1\times(n_s-2)}, \tag{A.28}$$

$$K_{bb}^{(1)} = k_0. \tag{A.29}$$

Force vector partition: $f_i^{(1)} = \mathbf{0}$, $f_b^{(1)} = 0$.

Computation of the Primal Schur complement

$K_{ii}^{(1)}$ is a tridiagonal matrix (diagonal entries $2k_0$, off-diagonal entries $-k_0$), whose inverse satisfies (verifiable directly):

$$\left(K_{ii}^{(1)}\right)^{-1}_{jk} = \frac{1}{k_0(n_s-1)} \min(j,k)\big(n_s - 1 - \max(j,k)\big), \quad 1 \le j,k \le n_s - 2. \tag{A.30}$$

Since $K_{bi}^{(1)}$ is nonzero only in the last column ($k = n_s - 2$), only the bottom-right entry of $(K_{ii}^{(1)})^{-1}$ is needed:

$$\left(K_{ii}^{(1)}\right)^{-1}_{n_s-2,\, n_s-2} = \frac{n_s - 2}{k_0(n_s - 1)}. \tag{A.31}$$

Substituting yields:

$$K_{bi}^{(1)}\left(K_{ii}^{(1)}\right)^{-1} K_{ib}^{(1)} = (-k_0)\cdot\frac{n_s - 2}{k_0(n_s - 1)}\cdot(-k_0) = k_0\frac{n_s - 2}{n_s - 1}. \tag{A.32}$$

Therefore:

$$\boxed{S_p^{(1)} = k_0 - k_0\frac{n_s - 2}{n_s - 1} = \frac{k_0}{n_s - 1} = \frac{k_0}{N_s} = \frac{ES}{H}.} \tag{A.33}$$

Right-hand side and rigid body mode

$$b_p^{(1)} = 0 - K_{bi}^{(1)}\left(K_{ii}^{(1)}\right)^{-1}\mathbf{0} = 0. \tag{A.34}$$

Since $S_p^{(1)} \neq 0$, subdomain (1) has **no rigid body mode**: $R_b^{(1)} = \emptyset$.

A.3.3 Block Partition and Schur Complement for Interior Subdomains ($2 \leq s \leq \mathcal{S} - 1$)

The interior subdomains are **floating subdomains** (no Dirichlet conditions), with both ends as interface nodes (local indices 1 and n_s), and interior nodes at indices 2 to $n_s - 1$ (a total of $n_s - 2$).

Original system

$$K^{(s)}u^{(s)} = f^{(s)}, \quad u^{(s)} = \begin{pmatrix} u_1 \\ \vdots \\ u_{n_s} \end{pmatrix}, \quad f^{(s)} = \mathbf{0}. \tag{A.35}$$

Interior/interface reordering

Reordering the degrees of freedom with interior nodes $(u_2, \cdots, u_{n_s-1})$ first and interface nodes (u_1, u_{n_s}) last, the corresponding matrix subblocks are:

$$K_{ii}^{(s)} = k_0 \begin{pmatrix} 2 & -1 & & \\ -1 & \ddots & \ddots & \\ & \ddots & 2 & -1 \\ & & -1 & 2 \end{pmatrix}_{(n_s-2)\times(n_s-2)}, \tag{A.36}$$

$$K_{ib}^{(s)} = k_0 \begin{pmatrix} -1 & 0 \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \\ 0 & -1 \end{pmatrix}_{(n_s-2)\times 2}, \tag{A.37}$$

$$K_{bi}^{(s)} = k_0 \begin{pmatrix} -1 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & -1 \end{pmatrix}_{2\times(n_s-2)}, \tag{A.38}$$

$$K_{bb}^{(s)} = k_0 I_2. \tag{A.39}$$

Force vector partition: $f_i^{(s)} = \mathbf{0}$, $f_b^{(s)} = \mathbf{0}$.

Computation of the Primal Schur complement

Let $m = n_s - 2$. From equation (A.30), the corner entries are:

$$\left(K_{ii}^{(s)}\right)^{-1}_{11} = \frac{m}{k_0(m+1)} = \frac{n_s - 2}{k_0(n_s - 1)}, \tag{A.40}$$

$$\left(K_{ii}^{(s)}\right)^{-1}_{mm} = \frac{n_s - 2}{k_0(n_s - 1)}, \tag{A.41}$$

$$\left(K_{ii}^{(s)}\right)^{-1}_{1m} = \left(K_{ii}^{(s)}\right)^{-1}_{m1} = \frac{1}{k_0(n_s - 1)}. \tag{A.42}$$

Therefore:

$$K_{bi}^{(s)}\left(K_{ii}^{(s)}\right)^{-1} K_{ib}^{(s)} = \frac{k_0}{n_s - 1}\begin{pmatrix} n_s - 2 & 1 \\ 1 & n_s - 2 \end{pmatrix}. \tag{A.43}$$

Hence:

$$\begin{aligned} S_p^{(s)} &= k_0 I_2 - \frac{k_0}{n_s - 1}\begin{pmatrix} n_s - 2 & 1 \\ 1 & n_s - 2 \end{pmatrix} \\ &= \frac{k_0}{n_s - 1}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} && \text{(A.44)} \\ &= \frac{k_0}{N_s}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} && \text{(A.45)} \\ &= \frac{ES}{H}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. && \text{(A.46)} \end{aligned}$$

$$\boxed{S_p^{(s)} = \frac{ES}{H}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.} \tag{A.47}$$

Right-hand side and rigid body mode

$$b_p^{(s)} = \mathbf{0} - K_{bi}^{(s)} \left(K_{ii}^{(s)}\right)^{-1} \mathbf{0} = \mathbf{0}. \tag{A.48}$$

Setting $S_p^{(s)} v = 0$ gives $v_1 = v_2$, so the rigid body mode is:

$$R_b^{(s)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{(global rigid body translation).} \tag{A.49}$$

A.3.4 Block Partition and Schur Complement for Subdomain ($\mathcal{S}$)

The right-end node of subdomain ($\mathcal{S}$) carries the applied force F_d. The interface node is the **first node** (local index 1), and the interior nodes are indices 2 to n_s (a total of $n_s - 1$).

Original system

$$K^{(\mathcal{S})} u^{(\mathcal{S})} = f^{(\mathcal{S})}, \quad u^{(\mathcal{S})} = \begin{pmatrix} u_1 \\ \vdots \\ u_{n_s} \end{pmatrix}, \quad f^{(\mathcal{S})} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ F_d \end{pmatrix}. \tag{A.50}$$

Interior/interface reordering

Reordering the degrees of freedom with interior nodes $(u_2, \cdots, u_{n_s})$ first and interface node (u_1) last, the sub-blocks are:

$$K_{ii}^{(\mathcal{S})} = k_0 \begin{pmatrix} 2 & -1 & & \\ -1 & \ddots & \ddots & \\ & \ddots & 2 & -1 \\ & & -1 & 1 \end{pmatrix}_{(n_s-1)\times(n_s-1)}, \tag{A.51}$$

$$K_{ib}^{(\mathcal{S})} = k_0 \begin{pmatrix} -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}_{(n_s-1)\times 1}, \quad K_{bi}^{(\mathcal{S})} = k_0 \begin{pmatrix} -1 & 0 & \cdots & 0 \end{pmatrix}_{1\times(n_s-1)}, \tag{A.52}$$

$$K_{bb}^{(\mathcal{S})} = k_0. \tag{A.53}$$

Force vector partition: $f_i^{(\mathcal{S})} = (0, \cdots, 0, F_d)^T$, $f_b^{(\mathcal{S})} = 0$.

Key: inverse of $K_{ii}^{(\mathcal{S})}$

This tridiagonal matrix corresponds to the 1D chain with a free boundary at the right end (bottom-right diagonal entry k_0); its inverse can be verified to be:

$$\left(K_{ii}^{(\mathcal{S})}\right)_{jk}^{-1} = \frac{\min(j,k)}{k_0}, \qquad 1 \le j, k \le n_s - 1. \tag{A.54}$$

In particular, the entire first row equals $1/k_0$, so $(K_{ii}^{(\mathcal{S})})_{11}^{-1} = 1/k_0$.

Computation of the Primal Schur complement

Since $K_{bi}^{(\mathcal{S})}$ is nonzero only in the first column and $K_{ib}^{(\mathcal{S})}$ is nonzero only in the first row, only $(K_{ii}^{(\mathcal{S})})_{11}^{-1}$ is needed:

$$K_{bi}^{(\mathcal{S})} \left(K_{ii}^{(\mathcal{S})}\right)^{-1} K_{ib}^{(\mathcal{S})} = (-k_0) \cdot \frac{1}{k_0} \cdot (-k_0) = k_0. \tag{A.55}$$

Therefore:

$$\boxed{S_p^{(\mathcal{S})} = k_0 - k_0 = 0.} \tag{A.56}$$

Computation of the right-hand side

Similarly, only $[(K_{ii}^{(\mathcal{S})})^{-1} f_i^{(\mathcal{S})}]_1$ is needed:

$$\left[\left(K_{ii}^{(\mathcal{S})}\right)^{-1} f_i^{(\mathcal{S})}\right]_1 = \sum_{k=1}^{n_s-1} \frac{\min(1,k)}{k_0} f_k = \frac{1}{k_0} \sum_{k=1}^{n_s-1} f_k = \frac{F_d}{k_0}. \tag{A.57}$$

Hence:

$$K_{bi}^{(\mathcal{S})} \left(K_{ii}^{(\mathcal{S})}\right)^{-1} f_i^{(\mathcal{S})} = (-k_0) \cdot \frac{F_d}{k_0} = -F_d, \tag{A.58}$$

$$\boxed{b_p^{(\mathcal{S})} = 0 - (-F_d) = F_d.} \tag{A.59}$$

Rigid body mode

Since $S_p^{(\mathcal{S})} = 0$, the equation $S_p^{(\mathcal{S})} R_b^{(\mathcal{S})} = 0$ holds for any scalar, so we may take:

$$R_b^{(\mathcal{S})} = 1. \tag{A.60}$$

A.3.5 Construction of the Primal Assembly Operator

There are $\mathcal{S} - 1$ global interfaces in total, denoted:

$$\Gamma_1, \Gamma_2, \ldots, \Gamma_{\mathcal{S}-1}.$$

The number of local interface degrees of freedom for each subdomain is:

- Subdomain (1): 1;
- Subdomain ($\mathcal{S}$): 1;
- Interior subdomains: 2 each.

The total number of local interface degrees of freedom is therefore $2\mathcal{S} - 2$.

The Primal assembly operator is defined as:

$$A^{(s)} \in \mathbb{R}^{(\mathcal{S}-1)\times n_b^{(s)}},$$

where:

- rows correspond to global interfaces Γ_ℓ;
- columns correspond to the local interface degrees of freedom of subdomain (s);
- if a local interface degree of freedom belongs to global interface Γ_ℓ, then $(A^{(s)})_{\ell j} = 1$, otherwise 0.

For the one-dimensional chain structure, the assembly operators have the explicit form:

$$A^{(1)} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \tag{A.61}$$

$$A^{(s)} = \begin{pmatrix} 0 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 0 & 1 \\ \vdots & \vdots \\ 0 & 0 \end{pmatrix}, \quad 2 \le s \le \mathcal{S} - 1, \tag{A.62}$$

$$A^{(\mathcal{S})} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}. \tag{A.63}$$

For the interior subdomains, the matrix has entries $(1, 0)$ and $(0, 1)$ in rows $s - 1$ and s respectively.

Define the concatenated block-diagonal matrix:

$$\mathbb{S}_p^{\diamondsuit} = \operatorname{diag}\big(S_p^{(1)}, S_p^{(2)}, \dots, S_p^{(\mathcal{S})}\big), \tag{A.64}$$

and the concatenated assembly operator:

$$\mathbb{A} = \big[A^{(1)}\ A^{(2)}\ \cdots\ A^{(\mathcal{S})}\big]. \tag{A.65}$$

The global Primal interface operator is then:

$$\boxed{\mathbf{S}_p = \mathbb{A}\,\mathbb{S}_p^{\diamondsuit}\,\mathbb{A}^T,} \qquad \boxed{\mathbf{B}_p = \mathbb{A}_p^{\diamondsuit}.} \tag{A.66}$$

The original finite element problem is thus equivalent to solving at the interface:

$$\mathbf{S}_p U_b = \mathbf{B}_p, \tag{A.67}$$

after which the interior degrees of freedom of each subdomain are recovered through local Dirichlet problems.

A.4 Relative Error

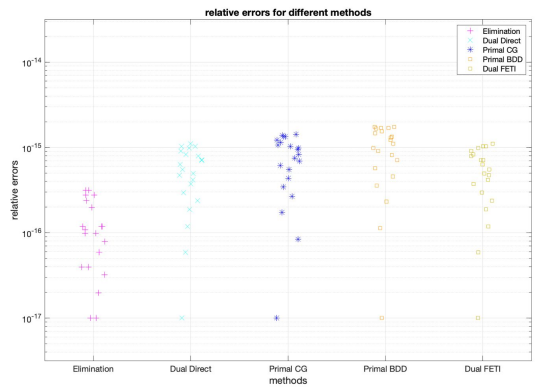


Figure 12: Relative errors for different methods.

B Primal Approach

B.1 Equivalence of Interface Residual and Global Residual

The global residual can be decomposed as a sum of local contributions from each subdomain:

$$r_{(k)} = f - K u_{(k)} = \sum_s \left[\begin{pmatrix} f_i^{(s)} \\ f_b^{(s)} \end{pmatrix} - \begin{pmatrix} K_{ii}^{(s)} & K_{ib}^{(s)} \\ K_{bi}^{(s)} & K_{bb}^{(s)} \end{pmatrix} \begin{pmatrix} u_{i(k)}^{(s)} \\ u_{b(k)}^{(s)} \end{pmatrix} \right]. \tag{B.1}$$

Since the interior degrees of freedom satisfy local equilibrium, the interior residual is zero, and therefore:

$$r_{(k)} = \sum_s \begin{pmatrix} 0 \\ r_{b(k)}^{(s)} \end{pmatrix}. \tag{B.2}$$

After assembly this yields:

$$\|r_{(k)}\| = \|\mathbf{r}_{b(k)}\|_\Gamma = \|\mathbf{b}_p - \mathbf{S}_p \mathbf{u}_{b(k)}\|_\Gamma. \tag{B.3}$$

The interface residual norm and the global residual norm are therefore fully equivalent.

B.2 Distributed Conjugate Gradient Algorithm

Algorithm 1 Distributed CG for the primal interface problem

Initialization: $\mathbf{u}_{b(0)} = \mathbf{0}$, $\mathbf{r}_{b(0)} = \mathbf{b}_p$, $\mathbf{d}_{b(0)} = \mathbf{r}_{b(0)}$
for $k = 0, 1, 2, \ldots$ **do**
  Distribute interface vector: $d^{\diamondsuit}_{b(k)} = \mathbb{A}^T \mathbf{d}_{b(k)}$
  for $s = 1$ to N_s **do**
    Solve local Dirichlet problem: $d^{(s)}_{i(k)} = -K^{(s)-1}_{ii} K^{(s)}_{ib} d^{(s)}_{b(k)}$
    Compute local contribution: $S^{(s)}_p d^{(s)}_{b(k)}$
  end for
  Assemble: $\mathbf{S}_p \mathbf{d}_{b(k)} = \mathbb{A}^{\diamondsuit}(S^{\diamondsuit}_p d^{\diamondsuit}_{b(k)})$
  $\alpha_k = \dfrac{\mathbf{r}^T_{b(k)} \mathbf{r}_{b(k)}}{\mathbf{d}^T_{b(k)} \mathbf{S}_p \mathbf{d}_{b(k)}}$
  $\mathbf{u}_{b(k+1)} = \mathbf{u}_{b(k)} + \alpha_k \mathbf{d}_{b(k)}$
  $\mathbf{r}_{b(k+1)} = \mathbf{r}_{b(k)} - \alpha_k \mathbf{S}_p \mathbf{d}_{b(k)}$
  $\beta_{k+1} = \dfrac{\mathbf{r}^T_{b(k+1)} \mathbf{r}_{b(k+1)}}{\mathbf{r}^T_{b(k)} \mathbf{r}_{b(k)}}$
  $\mathbf{d}_{b(k+1)} = \mathbf{r}_{b(k+1)} + \beta_{k+1} \mathbf{d}_{b(k)}$
  if $\|\mathbf{r}_{b(k+1)}\| / \|\mathbf{r}_{b(0)}\| \leq \varepsilon_{\text{tol}}$ **then**
    break
  end if
end for

C BDD Method

C.1 BDD-PCG Algorithm

Algorithm 2 BDD Preconditioned CG for the primal interface problem

Preprocessing: Compute the coarse space matrix G, assemble and factorize $G^T \mathbf{S}_p G$; assemble the Neumann preconditioner $\widetilde{\mathbf{S}}_p^{-1}$
Coarse problem initial solution:

$$\mathbf{u}_0 = G\big(G^T \mathbf{S}_p G\big)^{-1} G^T \mathbf{b}_p$$

Project initial residual:

$$\mathbf{r}_0 = P^T\big(\mathbf{b}_p - \mathbf{S}_p \mathbf{u}_0\big), \qquad P = I - G\big(G^T \mathbf{S}_p G\big)^{-1} G^T \mathbf{S}_p$$

if $\|\mathbf{r}_0\| \leq \varepsilon_{\text{tol}} \|\mathbf{b}_p\|$ **then**
  return $\mathbf{u}_0$ ▷ Coarse problem solution is sufficiently accurate; no CG iterations needed
end if
Precondition: $\mathbf{z}_0 = P\,\widetilde{\mathbf{S}}_p^{-1} \mathbf{r}_0, \quad \mathbf{d}_0 = \mathbf{z}_0, \quad \mathbf{u}^* = \mathbf{0}$
for $k = 0, 1, 2, \ldots$ **do**
  Distribute: $d^{\diamondsuit} = \mathbb{A}^T \mathbf{d}_k$
  for $s = 1$ to $\mathcal{S}$ **do**
    Local Dirichlet problem: $d^{(s)}_i = -\big(K^{(s)}_{ii}\big)^{-1} K^{(s)}_{ib} d^{(s)}_b$
    Local contribution: $S^{(s)}_p d^{(s)}_b$
  end for
  Assemble: $\mathbf{S}_p \mathbf{d}_k = \mathbb{A}^{\diamondsuit}(S^{\diamondsuit}_p d^{\diamondsuit})$
  Project: $\mathbf{p}_k = P^T \mathbf{S}_p \mathbf{d}_k$
  $\alpha_k = \dfrac{\mathbf{r}^T_k \mathbf{d}_k}{\mathbf{d}^T_k \mathbf{p}_k}$
  $\mathbf{u}^* \mathrel{+}= \alpha_k \mathbf{d}_k$
  $\mathbf{r}_{k+1} = \mathbf{r}_k - \alpha_k \mathbf{p}_k$
  $\mathbf{z}_{k+1} = P\,\widetilde{\mathbf{S}}_p^{-1} \mathbf{r}_{k+1}$
  if $\|\mathbf{r}_{k+1}\| / \|\mathbf{r}_0\| \leq \varepsilon_{\text{tol}}$ **then**
    break
  end if
  $\beta_{k+1} = \dfrac{\mathbf{r}^T_{k+1} \mathbf{z}_{k+1}}{\mathbf{r}^T_k \mathbf{z}_k}$ ▷ (full history used when orthogonalization is applied)
  $\mathbf{d}_{k+1} = \mathbf{z}_{k+1} + \beta_{k+1} \mathbf{d}_k$
end for
Final solution: $\mathbf{u}_b = \mathbf{u}_0 + \mathbf{u}^*$

D Dual Approach Appendix

D.1 Direct Solution Method

Local Subproblem for Each Subdomain

In the Dual method, displacement continuity across subdomains is **not imposed a priori**. Variations are taken separately with respect to $\underline{u}_E$ and $\underline{F}_{EE'}$ in the Lagrangian, and the resulting equations are then discretized. The local stiffness equation for subdomain s can be written in block form as:

$$\begin{pmatrix} k_{ii}^{(s)} & k_{ib}^{(s)} \\ k_{bi}^{(s)} & k_{bb}^{(s)} \end{pmatrix} \begin{pmatrix} u_i^{(s)} \\ u_b^{(s)} \end{pmatrix} = \begin{pmatrix} f_i^{(s)} \\ f_b^{(s)} \end{pmatrix} + \begin{pmatrix} 0 \\ \lambda_b^{(s)} \end{pmatrix}, \tag{D.1}$$

where $\lambda_b^{(s)}$ are the Lagrange multiplier forces acting on the interface nodes of subdomain s, arising from the interaction with neighboring subdomains.

A **unique** Lagrange multiplier $\underline{F}_{EE'} = -\underline{F}_{E'E}$ is introduced at each interface, so that the action-reaction principle is automatically satisfied:

$$F_{EE'} + F_{E'E} = 0. \tag{D.2}$$

In the finite element framework, the relationship between $\lambda_b^{(s)}$ and the global Lagrange multiplier λ_b is written as:

$$\lambda_b^{\diamondsuit} = \mathbb{A}^{\diamondsuit T} \lambda_b, \tag{D.3}$$

where the signs in $\mathbb{A}^{\diamondsuit}$ (the signed Boolean assembly operator) encode equation (D.2).

For each **floating substructure** (Neumann boundary conditions), $K^{(s)}$ is singular, and the necessary and sufficient condition for the system to have a solution (compatibility condition) is:

$$R^{(s)T}\left(f^{(s)} + \lambda^{(s)}\right) = 0, \tag{D.4}$$

i.e., the projection of the resultant of external loads and interface forces onto the rigid body mode direction is zero (rigid body equilibrium condition). In the 1D problem, each floating subdomain has one rigid body mode (constant translation), and equation (D.4) reduces to $\lambda_b^{(s)} + \lambda_b^{(s')} = 0$, which is equivalent to equation (D.2).

Dual Schur Complement Formulation of the Interface Problem

Eliminating the interior degrees of freedom from equation (D.1) yields the interface equation for each subdomain (following the same Schur complement procedure as in the Primal method):

$$S_p^{(s)} u_b^{(s)} = b_p^{(s)} + \lambda_b^{(s)}, \tag{D.5}$$

where $S_p^{(s)}$ and $b_p^{(s)}$ are defined identically to the Primal method (see Appendix D.1.1).

For floating substructures, $S_p^{(s)}$ is singular and generally not invertible. Introducing the generalized inverse $S_p^{(s)+}$, the general solution of the equation is:

$$u_b^{(s)} = S_p^{(s)+}\left(b_p^{(s)} + \lambda_b^{(s)}\right) + R_b^{(s)} \alpha_b^{(s)}, \tag{D.6}$$

where $R_b^{(s)}$ are the rigid body modes on the interface and $\alpha_b^{(s)}$ is the vector of undetermined amplitudes. Setting $S_d^{(s)} = S_p^{(s)+}$, we have:

$$u_b^{\diamondsuit} = S_d^{\diamondsuit}\left(b_p^{\diamondsuit} + \lambda_b^{\diamondsuit}\right) + R_b^{\diamondsuit} \alpha_b^{\diamondsuit}. \tag{D.7}$$

Substituting the above into the **displacement compatibility condition** $\mathbb{A}^{\diamondsuit} u_b^{\diamondsuit} = 0$ and using $\lambda_b^{\diamondsuit} = \mathbb{A}^{\diamondsuit T} \lambda_b$, one obtains the saddle-point system for λ_b and $\alpha_b^{\diamondsuit}$:

$$\boxed{\begin{pmatrix} S_d & G \\ G^T & 0 \end{pmatrix} \begin{pmatrix} \lambda_b \\ \alpha_b^{\diamondsuit} \end{pmatrix} = \begin{pmatrix} -b_d \\ -e^{\diamondsuit} \end{pmatrix}} \tag{D.8}$$

where the operators are defined as:

$$S_d = \mathbb{A}^{\diamondsuit} S_d^{\diamondsuit} \mathbb{A}^{\diamondsuit T}, \tag{D.9}$$

$$G = \mathbb{A}^{\diamondsuit} R_b^{\diamondsuit}, \tag{D.10}$$

$$b_d = \mathbb{A}^{\diamondsuit} S_d^{\diamondsuit} b_p^{\diamondsuit}, \tag{D.11}$$

$$e^{\diamondsuit} = R_b^{\diamondsuit T} b_p^{\diamondsuit}. \tag{D.12}$$

Here S_d is the **dual Schur complement**. The complete derivation of system (D.8) is given in Appendix D.1.2.

D.1.1 Definition of the Local Schur Complement

For subdomain s, eliminating the interior degrees of freedom yields the local Primal Schur complement and its right-hand side:

$$S_p^{(s)} = k_{bb}^{(s)} - k_{bi}^{(s)} \left(k_{ii}^{(s)}\right)^{-1} k_{ib}^{(s)}, \tag{D.13}$$

$$b_p^{(s)} = f_b^{(s)} - k_{bi}^{(s)} \left(k_{ii}^{(s)}\right)^{-1} f_i^{(s)}. \tag{D.14}$$

The local Dual Schur complement is defined as the Moore–Penrose pseudoinverse of $S_p^{(s)}$:

$$S_d^{(s)} = S_p^{(s)+}. \tag{D.15}$$

D.1.2 Complete Derivation of the Dual Saddle-Point System

Step 1: Introduction of Lagrange multipliers. A unique Lagrange multiplier $F_{EE'} = -F_{E'E}$ is introduced at each interface $\Gamma_{EE'}$, so that the action-reaction principle $F_{EE'} + F_{E'E} = 0$ is automatically satisfied. After finite element discretization, the force acting on the interface nodes of each subdomain is:

$$\lambda_b^\diamond = \mathbb{A}^{\diamond T} \lambda_b. \tag{D.16}$$

Step 2: Condensation of the local problems. The local equations for each subdomain are:

$$\begin{pmatrix} k_{ii}^{(s)} & k_{ib}^{(s)} \\ k_{bi}^{(s)} & k_{bb}^{(s)} \end{pmatrix} \begin{pmatrix} u_i^{(s)} \\ u_b^{(s)} \end{pmatrix} = \begin{pmatrix} f_i^{(s)} \\ f_b^{(s)} \end{pmatrix} + \begin{pmatrix} 0 \\ \lambda_b^{(s)} \end{pmatrix}. \tag{D.17}$$

Eliminating $u_i^{(s)}$ from the first row (assuming $k_{ii}^{(s)}$ is invertible):

$$u_i^{(s)} = \left(k_{ii}^{(s)}\right)^{-1} \left(f_i^{(s)} - k_{ib}^{(s)} u_b^{(s)}\right). \tag{D.18}$$

Substituting into the second row yields:

$$S_p^{(s)} u_b^{(s)} = b_p^{(s)} + \lambda_b^{(s)}. \tag{D.19}$$

Step 3: General solution for floating substructures. For floating substructures, $S_p^{(s)}$ is singular. The solvability condition for equation (D.19) is (equilibrium):

$$R_b^{(s)T} \left(b_p^{(s)} + \lambda_b^{(s)}\right) = 0. \tag{D.20}$$

When condition (D.20) is satisfied, the general solution is:

$$u_b^{(s)} = \underbrace{S_d^{(s)} \left(b_p^{(s)} + \lambda_b^{(s)}\right)}_{\text{particular solution (pseudoinverse)}} + \underbrace{R_b^{(s)} \alpha_b^{(s)}}_{\text{rigid body mode component}}. \tag{D.21}$$

Writing all subdomains in block-diagonal form:

$$u_b^\diamond = S_d^\diamond \left(b_p^\diamond + \lambda_b^\diamond\right) + R_b^\diamond \alpha_b^\diamond. \tag{D.22}$$

Step 4: Substitution into the displacement compatibility condition. The displacement compatibility condition (zero displacement jump at the interface) is:

$$\mathbb{A}^\diamond u_b^\diamond = 0. \tag{D.23}$$

Substituting equations (D.22) and (D.16) into equation (D.23):

$$\begin{aligned} &\mathbb{A}^\diamond \left[S_d^\diamond \left(b_p^\diamond + \mathbb{A}^{\diamond T} \lambda_b\right) + R_b^\diamond \alpha_b^\diamond\right] = 0 \\ \implies \quad &\underbrace{\mathbb{A}^\diamond S_d^\diamond \mathbb{A}^{\diamond T}}_{S_d} \lambda_b + \underbrace{\mathbb{A}^\diamond R_b^\diamond}_{G} \alpha_b^\diamond = -\underbrace{\mathbb{A}^\diamond S_d^\diamond b_p^\diamond}_{b_d}. \end{aligned} \tag{D.24}$$

Step 5: Substitution into the rigid body equilibrium constraint. Substituting equation (D.16) into the equilibrium condition (D.20) and writing in global block form:

$$R_b^{\diamond T} \left(b_p^\diamond + \mathbb{A}^{\diamond T} \lambda_b\right) = 0 \quad \implies \quad \underbrace{R_b^{\diamond T} \mathbb{A}^{\diamond T}}_{G^T} \lambda_b = -\underbrace{R_b^{\diamond T} b_p^\diamond}_{e^\diamond}. \tag{D.25}$$

Step 6: Dual saddle-point system. Combining equations (D.24) and (D.25):

$$\begin{pmatrix} S_d & G \\ G^T & 0 \end{pmatrix} \begin{pmatrix} \lambda_b \\ \alpha_b^\diamond \end{pmatrix} = \begin{pmatrix} -b_d \\ -e^\diamond \end{pmatrix}, \tag{D.26}$$

where:

$$S_d = \mathbb{A}^\diamond S_d^\diamond \mathbb{A}^{\diamond T}, \quad G = \mathbb{A}^\diamond R_b^\diamond, \tag{D.27}$$

$$b_d = \mathbb{A}^\diamond S_d^\diamond b_p^\diamond, \quad e^\diamond = R_b^{\diamond T} b_p^\diamond. \tag{D.28}$$

Symmetric positive semi-definiteness of S_d. For any vector μ:

$$\mu^T S_d \mu = \mu^T \mathbb{A}^\diamond S_d^\diamond \mathbb{A}^{\diamond T} \mu = \left(\mathbb{A}^{\diamond T} \mu\right)^T S_d^\diamond \left(\mathbb{A}^{\diamond T} \mu\right) \geq 0, \tag{D.29}$$

since each $S_d^{(s)} = S_p^{(s)+}$ is positive semi-definite. Hence S_d is symmetric positive semi-definite.

D.2 Displacement Recovery Algorithm

After solving the saddle-point system (D.26) for $(\lambda_b, \alpha_b^\diamond)$, the displacements are recovered subdomain by subdomain as follows:

Algorithm 3 Dual approach: displacement recovery

```
for s = 1 to S do
    Distribute interface forces: λ_b^(s) ← A^(s)T λ_b
    Compute interface displacements (particular solution plus rigid body mode): u_b^(s) ← S_d^(s)(b_p^(s) + λ_b^(s)) + R_b^(s) α_b^(s)
    Recover interior displacements: u_i^(s) ← (k_ii^(s))^-1 (f_i^(s) − k_ib^(s) u_b^(s))
end for
```

D.3 Explicit Construction of the $\mathbb{A}^\diamond$ Operator for the 1D Problem

For a chain decomposition of $\mathcal{S}$ equal-size subdomains, there are $\mathcal{S}-1$ global Lagrange multipliers in total. Sign convention: at the j-th interface (between subdomains j and $j+1$), the right-end node of subdomain j contributes $+\lambda_j$, and the left-end node of subdomain $j+1$ contributes $-\lambda_j$.

The distribution relation $\lambda_b^\diamond = \mathbb{A}^{\diamond T}\lambda_b$ expands to:

- Subdomain 1 (right end): ${\lambda_b^{(1)}}_{\text{right}} = +\lambda_1$
- Subdomain s (interior, $2 \le s \le \mathcal{S}-1$): ${\lambda_b^{(s)}}_{\text{left}} = -\lambda_{s-1}$, ${\lambda_b^{(s)}}_{\text{right}} = +\lambda_s$
- Subdomain $\mathcal{S}$ (left end): ${\lambda_b^{(\mathcal{S})}}_{\text{left}} = -\lambda_{\mathcal{S}-1}$

It can be verified that ${\lambda_b^{(s)}}_{\text{right}} + {\lambda_b^{(s+1)}}_{\text{left}} = \lambda_s - \lambda_s = 0$, i.e., the action-reaction principle (D.2) is automatically satisfied.

The compatibility condition $\mathbb{A}^\diamond u_b^\diamond = 0$ expands to:

$$u_{\text{right}}^{(j)} - u_{\text{left}}^{(j+1)} = 0, \quad j = 1, \dots, \mathcal{S}-1. \tag{D.30}$$

D.4 FETI Method: Projected Preconditioned Conjugate Gradient Algorithm

The following algorithm is presented in the form of Algorithm 2 from the reference, with notation consistent with the main text (b subscripts and global Lagrange multiplier subscripts are omitted for brevity).

Algorithm 4 Dual Schur complement with conjugate gradient (FETI)

Initialization:

1: $P = I - QG(G^TQG)^{-1}G^T$

2: $\lambda_{(0)} = -QG(G^TQG)^{-1}e^\diamond$

3: $r_{(0)} = P^T(-b_d - S_d\,\lambda_{(0)})$

4: $z_{(0)} = P\,\widetilde{S}_d^{-1}\,r_{(0)}$ and $d_{(0)} = z_{(0)}$

loop For $i = 0$ to m **do**

5: Set: $p_{(i)} = P^T S_d\, d_{(i)}$

6: Compute optimal step: $\alpha_{(i)} = \dfrac{r_{(i)}^T\, d_{(i)}}{d_{(i)}^T\, p_{(i)}}$

7: Compute iterate: $\lambda_{(i+1)} = \lambda_{(i)} + \alpha_{(i)}\, d_{(i)}$

8: Compute residual: $r_{(i+1)} = r_{(i)} - \alpha_{(i)}\, p_{(i)}$

9: Set: $z_{(i+1)} = P\,\widetilde{S}_d^{-1}\,r_{(i+1)}$

10: Reorthogonalization: For $j = 0$ to i **do**, $\beta_{(i)}^j = -\dfrac{z_{(i+1)}^T\, p_{(j)}}{d_{(j)}^T\, p_{(j)}}$

11: Update search direction: $d_{(i+1)} = z_{(i+1)} + \displaystyle\sum_{j=1}^{i} \beta_{(i)}^j\, d_{(i)}$

end loop

Post-processing:

12: $\alpha_b^\diamond = (G^TQG)^{-1}G^TQ\big(-b_d - S_d\,\lambda_{(m)}\big)$

13: $u_b^\diamond = S_p^{+\diamond}\big(b_p^\diamond + \mathbb{A}^{\diamond T}\lambda_{(m)}\big) + R_b^\diamond\,\alpha_b^\diamond$

Choice of Q. For structures with uniform material properties, $Q = I$ is taken, in which case the projection operator P is an orthogonal projection (Hermitian) satisfying $PP^T = I$. This choice is adopted in the present problem.

Distributed computation of matrix-vector products. The computation of $S_d d_{(i)}$ in Step 1 is carried out in a distributed manner:

$$S_d\, d = \mathbb{A}^\diamond S_d^\diamond \mathbb{A}^{\diamond T} d, \tag{D.31}$$

where $\mathbb{A}^{\diamond T} d$ is the distribution step (broadcasting the vector to each subdomain), $S_d^\diamond \cdot$ denotes the local matrix-vector multiplication within each subdomain, and $\mathbb{A}^\diamond \cdot$ is the assembly step (signed summation). Only interface vectors are exchanged between subdomains, with a communication volume of $\mathcal{O}(\mathcal{S}-1)$.

Distributed computation of the Dirichlet preconditioner. Similarly, $\widetilde{S}_d^{-1} r$ is computed as:

$$\widetilde{S}_d^{-1}\, r = \mathbb{A}^\diamond S_p^\diamond \mathbb{A}^{\diamond T} r, \tag{D.32}$$

i.e., using $S_p^{(s)}$ (rather than $S_d^{(s)}$) for the local matrix-vector multiplication, again without assembling the global matrix.

E Mixed Methods

E.1 Monoscale LATIN-based Mixed DDM Algorithm

The following is the original algorithm from the reference (Algorithms.pdf, Algorithm 4), reproduced without modification.

Algorithm 5 Monoscale LATIN-based mixed DDM

Initialization:

- For each substructure $E \in \mathbf{E}$:
  - ☐ Computation of the stiffness matrices: $[K_E]$ and $[k_E]$,
  - ☐ Assembly, factorization of $[K_E] + [k_E]$.
- Initialization of $\widehat{s}_{-1/2}$
  - ☐ Choose s_{-1} (generally set to 0)
  - ☐ Preliminary local stage $\rightarrow \widehat{s}_{-1/2}$

loop For $n = 0$ to m **do**

1. Linear stage: Computation of s_n
   - ☐ Computation of $u_E^{(n)}$ (and $W_E^{(n)}$) for each substructure $E \in \mathbf{E}$:

$$\rightarrow \quad \text{Solve } \left([K_E] + [k_E^-]\right)[u_E^{(n)}] = [\widehat{F}_{E,d}]$$

   - ☐ Determination of $F_E^{(n)}$ with search direction E^-:

$$\rightarrow \quad F_E^{(n)} = \widehat{F}_E^{(n-1/2)} - k^-\left(W_E^{(n)} - \widehat{W}_E^{(n-1/2)}\right)$$

2. Optional relaxation stage: $s_n \leftarrow \mu s_n + (1-\mu)s_{n-1}$ where $\mu = 0.8$ in practice.
3. Local stage: Computation of $\widehat{s}_{n+1/2}$
   - ☐ Computation of $\widehat{W}_{EE'}^{(n+1/2)}$ and $\widehat{F}_{EE'}^{(n+1/2)}$ for each interface $\Gamma_{EE'}$. For a perfect interface:

$$\left|\begin{aligned}
\widehat{W}_{EE'}^{(n+1/2)} &= \widehat{W}_{E'E}^{(n+1/2)} \\
&= \tfrac{1}{2}\left(W_{EE'}^{(n)} + W_{E'E}^{(n)}\right) \\
&\quad - \tfrac{(k^+)^{-1}}{2}\left(F_{EE'}^{(n)} + F_{E'E}^{(n)}\right), \\
\widehat{F}_{EE'}^{(n+1/2)} &= -\widehat{F}_{E'E}^{(n+1/2)} \\
&= F_{EE'}^{(n)} \\
&\quad + k^+\left(\widehat{W}_{EE'}^{(n+1/2)} - W_{EE'}^{(n)}\right).
\end{aligned}\right.$$

4. Convergence criterion: Computation of η

end loop

E.2 Convergence Error Indicator

The convergence of the LATIN method is measured through the conjugacy of the search directions:

$$\eta^2 = \frac{\sum_{E\in\mathbf{E}} \Big[\|W_E - \widehat{W}_E\|^2_{W,k} + \|F_E - \widehat{F}_E\|^2_{F,k} \Big]}{\sum_{E\in\mathbf{E}} \Big[\|W_E\|^2_{W,k} + \|\widehat{W}_E\|^2_{W,k} + \|F_E\|^2_{F,k} + \|\widehat{F}_E\|^2_{F,k} \Big]}, \tag{E.1}$$

where

$$\|W\|^2_{W,k} = \int_{\partial\Omega_E} k\, W\cdot W \,\mathrm{d}\Gamma, \qquad \|F\|^2_{F,k} = \int_{\partial\Omega_E} F\cdot k^{-1} F \,\mathrm{d}\Gamma. \tag{E.2}$$

For the 1D bar problem, each interface reduces to a single point, and the above integrals degenerate to scalar products:

$$\eta^2 = \frac{\sum_j \Big[k\,(W_j - \widehat{W}_j)^2 + k^{-1}(F_j - \widehat{F}_j)^2 \Big]}{\sum_j \Big[k\, W_j^2 + k\, \widehat{W}_j^2 + k^{-1} F_j^2 + k^{-1} \widehat{F}_j^2 \Big]}, \tag{E.3}$$

where the summation runs over all interface nodes, and k is the search direction parameter (with $k^+ = k^- = k$).

E.3 Remarks on the 1D Discretization of the Search Direction Operator

In the finite element discretization of the 1D bar, substructure E contains several nodes, and the interface nodes are the endpoints shared with neighboring substructures. The discretized form of the linear stage equation (6.2) reads:

$$\big(K_E + k^- I_b\big)\, u_E^{(n)} = f_{E,d}^{(n)}, \tag{E.4}$$

where I_b is a diagonal matrix with ones only at the interface degrees of freedom, and the right-hand side is

$$f_{E,d}^{(n)} = f_E + \widehat{F}_E^{(n-1/2)} + k^- \widehat{W}_E^{(n-1/2)}. \tag{E.5}$$

This matrix remains unchanged throughout the iterations (only the right-hand side is updated), so a single LU factorization can be performed during initialization, and only forward/backward substitutions are required at each iteration.

E.4 Multiscale LATIN-based Mixed DDM Algorithm

The following is the original algorithm from the course reference (Latin_based_multiscale_DDM.pdf, Algorithm 5), reproduced without modification.

Algorithm 6 Multiscale LATIN-based mixed DDM

Initialization:

- For each substructure $E \in \mathbf{E}$:
  - ☐ Computation of the stiffness matrices: $[K_E]$ and $[k_E]$,
  - ☐ Assembly, factorization of $[K_E] + [k_E]$.
  - ☐ Determination of the homogenized operator: $[\mathbf{L}_E^F]$.
- Assembly and factorization of the homogenized operator $[\mathbf{L}^F]_{e^M}$
- Initialization of $\widehat{s}_{-1/2}$
  - ☐ Choose s_{-1} (generally set to 0)
  - ☐ Preliminary local stage $\rightarrow \widehat{s}_{-1/2}$

loop For $n = 0$ to m **do**

1. Linear stage: Computation of s_n
   - ☐ Solve microproblem for known quantities – computation of $\breve{u}_E^{(n)}$ (and $\breve{W}_E^{(n)}$) for each substructure $E \in \mathbf{E}$:

     $$\rightarrow \quad \text{Solve } \left([K_E] + [k_E^-]\right) [\breve{u}_E^{(n)}] = [\widehat{F}_{E,d}]$$

     $$\breve{F}_E^{(n)} = \widehat{F}_E^{(n-1/2)} - k^- \left(\breve{W}_E^{(n)} - \widehat{W}_E^{(n-1/2)}\right) \quad \text{(search direction)}.$$

   - ☐ Solve macroproblem – computation of Lagrange multiplier $\widetilde{W}^M$:

     $$\rightarrow \quad [\mathbf{L}^F]_{e^M} [\widetilde{W}^M]_{e^M} = [\widehat{F}_d^M]_{e^M} + [F_d^M]_{e^M}$$

   - ☐ Solve microproblem for Lagrange multiplier loading – computation of $\tilde{u}_E^{(n)}$ (and $\tilde{W}_E^{(n)}$) for each substructure $E \in \mathbf{E}$:

     $$\rightarrow \quad \text{Solve } \left([K_E] + [k_E^-]\right) [\tilde{u}_E^{(n)}] = [k\widetilde{W}_E^M]$$

     $$\rightarrow \quad \tilde{F}_E^{(n)} = \widehat{F}_E^{(n-1/2)} - k^- \left(\tilde{W}_E^{(n)} - \widetilde{W}_E^M\right) \quad \text{(search direction)}$$

     $$\rightarrow \quad \text{Compute } u_E = \breve{u}_E + \tilde{u}_E \text{ and } F_E^{(n)} = \breve{F}_E^{(n)} + \tilde{F}_E^{(n)}$$

2. Optional relaxation stage: $s_n \leftarrow \mu s_n + (1-\mu) s_{n-1}$ where $\mu = 0.8$ in practice.
3. Local stage: Computation of $\widehat{s}_{n+1/2}$
   - ☐ Computation of $\widehat{W}_{EE'}^{(n+1/2)}$ and $\widehat{F}_{EE'}^{(n+1/2)}$ for each interface $\Gamma_{EE'}$ (same as monoscale Algorithm 4).
4. Convergence criterion: Computation of η

end loop

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